

SX1302 868M LoRaWAN Gateway

From Waveshare Wiki

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Introduction

This is a communication HAT for Raspberry Pi with LoRaWAN protocol support, which incorporates the core LoRa gateway module SX1302 868M LoRaWAN Gateway. It has lots of advantages including long-range transmission, concurrent communication, large capacity node, and high receiving sensitivity. There are L76B GNSS module and Mini-PCIe socket on the baseboard for easy integration, along with our rich development resources, you can attach it to Raspberry Pi and build the LoRa network fast.

SX1302 868M LoRaWAN Gateway



(<https://www.waveshare.com/sx1302-868m-lorawan-gateway-b.htm>)

Features

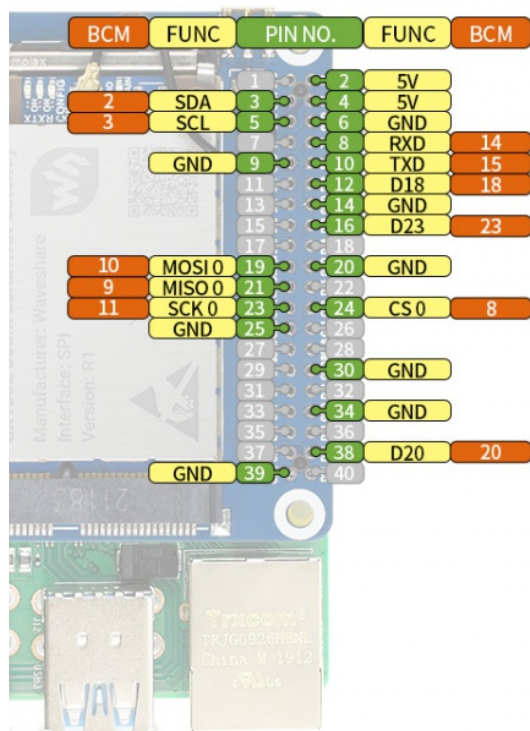
- LoRaWAN Gateway Module features:
 - Integrates Semtech SX1302/3 normal band and SX1250 radio RF frond-end chip.
 - Onboard PA and LNA, features +26dBm emit power and -141dBm high sensitivity receiving gain.
 - The SX1303 supports Fine Timestamp and network positioning based on the time difference of arrival (TDOA).
 - 52-pin Mini-PCIe socket for easy integration into various embedded systems.
 - Onboard 4 LED indicators for module operating status.
 - Comes with development resources and manual (example in C).
- HAT features:
 - Standard Raspberry Pi 40PIN GPIO extension header, supports Raspberry Pi series boards.
 - Incorporate the L76K module with GPS/BD support, and provide accurate clock and location info for the gateway module.
 - Comes with online development resources and manual (example in C).

Specification

	LORA GATEWAY	GNSS MODULE
Power supply	5V	
Fequency range	EU868 (863-870MHz)	GPS L1 (1575.42MHz) BD2 B1 (1561.098MHz)
Positioning accuracy	-	< 2.5m CEP
Modulation	LoRa/(G)FSK	-
Emit power	> 26dBm@5V	-

Receiving sensitivity	-141dBm@125KHz/SF12 -121dBm@125KHz/SF5	acquisition: -148dBm tracking: -163dBm re-acquisition: -160dBm
Overall power consumption	emitting: 710mA@5V GPS on receiving: 99mA@5V GPS on sleep: 41mA@5V GPS off	
Communication bus	SPI, I2C	UART
External connector	Mini-PCIe	-
Operating temperature	-40~85°C	
Dimensions	50.95 x 30mm	

Pinout



GNSS Module	
5V	5V power supply
GND	Ground
RXD	L76B module UART RX
TXD	L76B module UART TX

LoRa Gateway	
SDA	Module I2C SDA
SCL	Module I2C SCL
MOSI 0	Module SPI data output
MISO 0	Module SPI data input
SCK 0	Module SPI clock input
CS 0	Module SPI chip select
D18	Module radio power on/off
D20	Module GPIO6 multiplex pin
D23	Module reset pin

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File:SX1302-LoRaWAN-Gateway-HAT-details-inter.jpg)

LoRa & LoRaWAN

What is LoRa ?

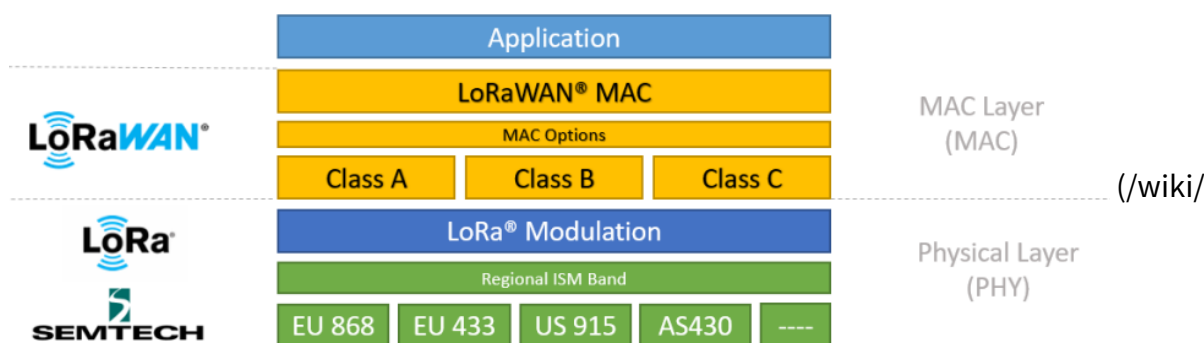
[Semtech (<https://www.semtech.com/>)]'s LoRa is a long-distance, low-power wireless platform for the Internet of Things (IoT), which generally refers to radio frequency chips using LoRa technology. The main features are as follows:

- The spread spectrum modulation technology used by LoRa (abbreviation of long-range) is derived from chirp spread spectrum (CSS) technology, which is one of the long-distance wireless transmission technology and LPWAN communication technology. At present, LoRa mainly operates in the ISM frequency band, mainly including 433, 868, 915 MHz, etc.

- LoRa technology integrates technologies such as digital spread spectrum, digital signal processing, and forward error correction coding, which greatly improves the performance of long-distance communication. LoRa's link budget is better than any other standardized communication technology. The main factors that determine the distance in a given environment
- LoRa RF chips mainly include SX127X series, SX126X series, SX130X series, of which SX127X, SX126X series are used for LoRa nodes, and SX130X is used for LoRa gateways. For details, please refer to [Semtech (<https://www.semtech.com/>)]'s product list

What is LoRaWAN ?

- [LoRaWAN (<https://loro-alliance.org/about-lorawan/>)] is a low-power wide area network open protocol based on LoRa radio modulation technology. Aims to wirelessly connect battery-powered "things" to the Internet in a regional, national, or global network and targets key Internet of Things (IoT) requirements such as bi-directional communications, end-to-end security, mobility, and localized services. Node wireless connection to the Internet has access authentication, which is equivalent to the establishment of the encrypted communication channel between node and server. Access details refer to [document] and [source code (<https://github.com/Lora-net/LoRaMac-node>)]]. LoRaWAN protocol level is shown in the figure below.
- The Class A/B/C node devices in the MAC layer basically cover all the application scenarios of the Internet of Things. The differences among the three nodes lie in the different time slots for receiving and receiving nodes.
- EU868 and AS430 in the Modulation layer show that frequency band parameters are different in different countries. Please click the reference [link (https://loro-alliance.org/wp-content/uploads/2019/11/rp_2-1.0.0_final_release.pdf)] for regional parameters.

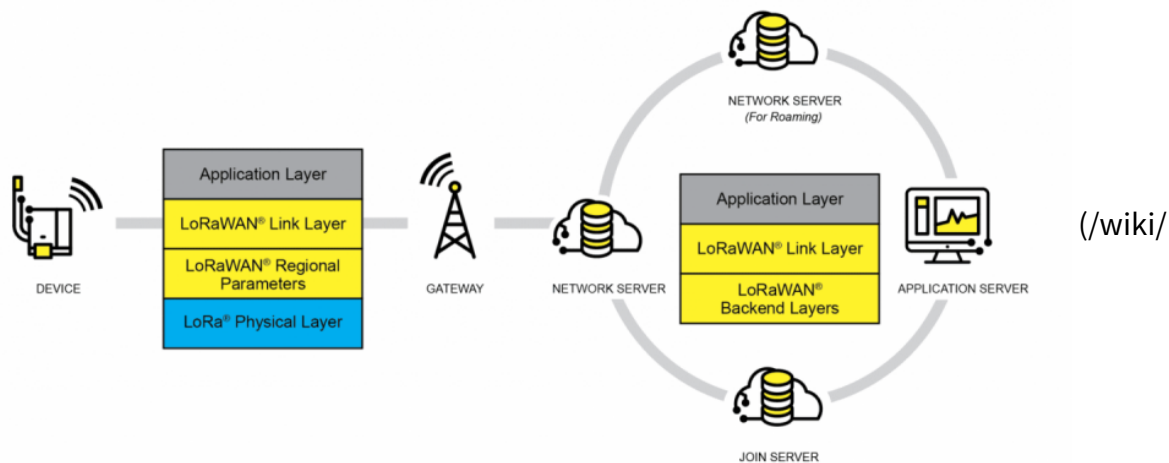


File: SX1262-LoRa-HAT-021.png

To achieve LoRaWAN network coverage in cities or other areas, it needs to be composed of four parts: node (LoRa node radio frequency chip), gateway (or base station, LoRa gateway radio frequency chip), server, and cloud, as shown in the following figure.

- The DEVICE (node device) needs to initiate a network access request packet to the GATEWAY (gateway) and then to the server. After the authentication is passed, it can send and receive application data with the server normally.
- GATEWAY (gateway) can communicate with the server through a wired network, 3/4/5G wireless network
- The main operators on the server side are [TTN (<https://eu1.cloud.thethings.network/console/>)], etc. For building cloud services by yourself, please refer to [lorawan-stack (<https://github.co>)]

m/TheThingsNetwork/lorawan-stack)], [chirpstack (<https://www.chirpstack.io/>)]



File:1350px-SX1268-LoRa-HAT-021.png)

Application

LoRa devices and networks such as LoRaWAN enable smart IoT applications to help solve the planet's formidable challenges in energy management, natural resource reduction, pollution control, infrastructure efficiency, disaster prevention, and more. Semtech's LoRa devices have achieved hundreds of successful use cases in smart cities, homes and buildings, communities, metrology, supply chain and logistics, agriculture, and more. LoRa networks have covered hundreds of millions of devices in more than 100 countries and are committed to a smarter planet. The examples here are based on the TTS (<https://eu1.cloud.thethings.network/oauth/login>) The THINGS STACK) and the [https://github.com/Lora-net/sx1302_hal (https://github.com/Lora-net/sx1302_hal) Semtech SX1203 libraries).

If you want to create the cloud server yourself, please refer to LoRaWan-Stack (<https://github.com/TheThingsNetwork/lorawan-stack>) and chirpstack (<https://www.chirpstack.io/>).

Hardware Connection

Attach the LoRa HAT on the Raspberry Pi and set the GPS antenna in an open area where the sky is visible.

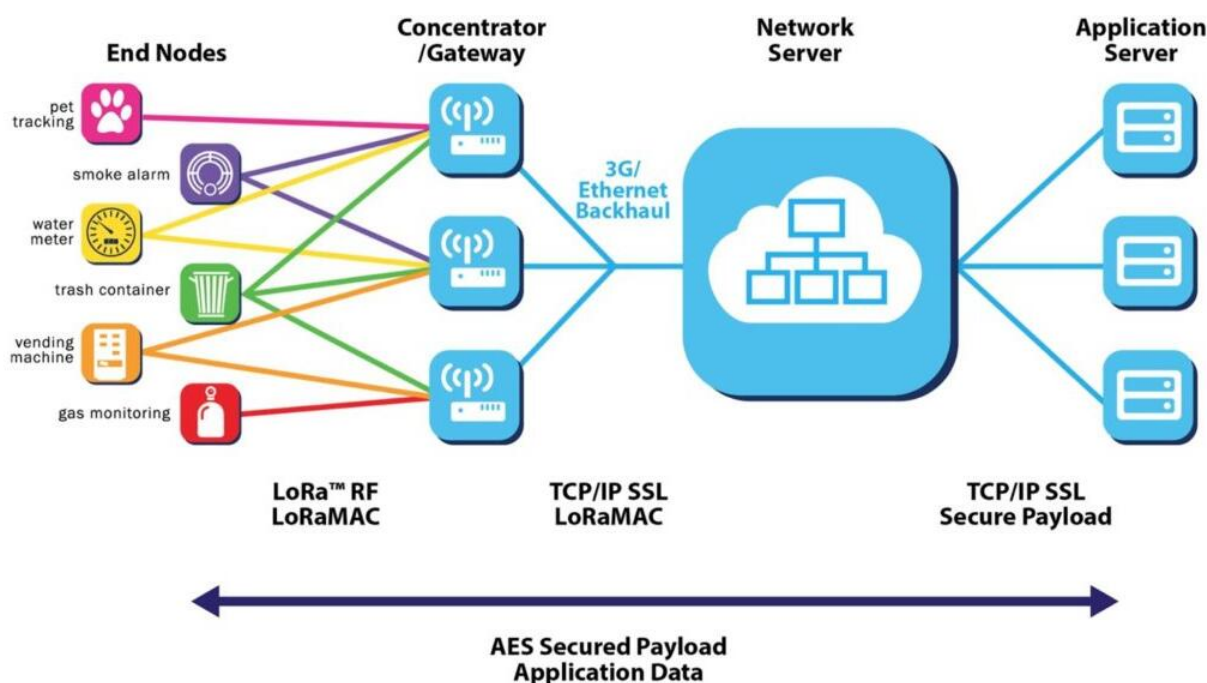


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File: SX1302_LoRaWAN_Gateway_HAT_001.jpg)

Network Instructions

- MCU runs LoRaWAN protocol and connects to Pico-LoRa-SX1262-868M as a node device (End Node), and through SX1302 LoRa Gateway upload data to TTS (<https://eu1.cloud.thethings.net/work/console/>) server, or receive data from TTS (<https://eu1.cloud.thethings.net/work/console/>) server data, the specific network topology is as shown in the picture below:



(/wiki/File:Pico-LoRa-SX1262-868M_005.jpg)

- Please select the applicable frequency band according to the local radio management regulations. Pico-LoRa-SX1262-868M is suitable for use in EU868 regional countries. The frequency band distribution diagram is as follows. For the specific table, click link (https://loralliance.org/wp-content/uploads/2019/11/rp_2-1.0.0_final_release.pdf) view or visit LoRa

Alliance (<https://lora-alliance.org/>).

ISO 3166-1 Country name (Code alpha-2)	Band / channels	Channel Plan	LoRaWAN Certified devices with Regulatory Type Approval
Afghanistan (AF)			
Aland Islands (AX)	433.05 - 434.79 MHz	EU433	
	863 - 870 MHz	EU863-870	
Albania (AL)	433.05 - 434.79 MHz	EU433	
	863 - 873 MHz	EU863-870	
	915 - 918 MHz		
<i>Algeria (DZ)</i>	433.05 – 434.79 MHz	EU433	
	870-876MHz		
	880-885MHz		
	915 – 921 MHz		
	925 – 926 MHz		
American Samoa (AS)	902 - 928 MHz	US902-928 AU915-928	X, _
Andorra (AD)	433.05 – 434.79 MHz	EU433	
	863 – 870 MHz	EU863-870	
Angola (AO)			
Anguilla (AI)	915 - 928 MHz ¹	AU915-928 AS923	
Antarctica (AQ)			
Antigua and Barbuda (AG)			
Argentina (AR)	915 - 928 MHz ¹	AU915-928	
Armenia (AM)	863 – 870 MHz	EU863-870	
	433.05 – 434.79 MHz	EU433	
Aruba (AW)			
Australia (AU)	915 - 928 MHz	AS923 AU915-928	X, X
<i>Austria (AT)</i>	433.05 - 434.79 MHz	EU433	

(/wiki/File:Pico-LoRa-SX1262-868M_004.jpg)

Install Libraries

Open the terminal of the Raspberry Pi and install libraries with the following commands:

Raspberry Pi 4B

```
sudo apt update
sudo apt install git
cd ~/Documents/
git clone https://github.com/Lora-net/sx1302_hal.git
cd sx1302_hal
make clean all
make all
cp tools/reset_lgw.sh util_chip_id/
cp tools/reset_lgw.sh packet_forwarder/
```

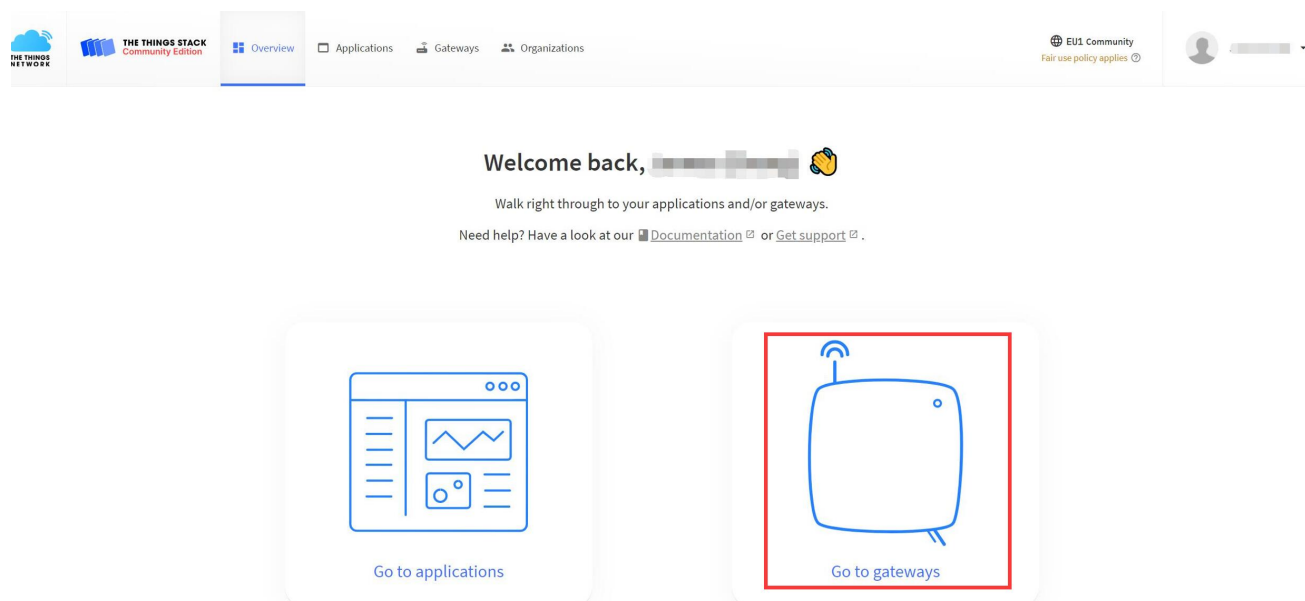
Raspberry Pi 5

```
wget https://files.waveshare.com/wiki/SX130X/demo/PI5/sx130x_hal_rpi5.zip
sudo unzip sx130x_hal_rpi5.zip
cd sx1302_hal_rpi5-master/
make clean all
make all
cp tools/reset_lgw.sh util_chip_id/ cp tools/reset_lgw.sh packet_forwarder/
cd sx1302_hal_rpi5-master/util_chip_id/
sudo ./chip_id
```

Connect TTS

- Sign up for a TTS account and login. (<https://id.thethingsnetwork.org/oidc/interaction/l3xePHpVtpa1NAPBs72wj>)
- Get the gateway EUI and add the gateway device to TTS, please choose the correct frequency according to the local area.

```
# get sx1302 EUI
cd ~/Documents/sx1302_hal/util_chip_id/
./chip_id
```



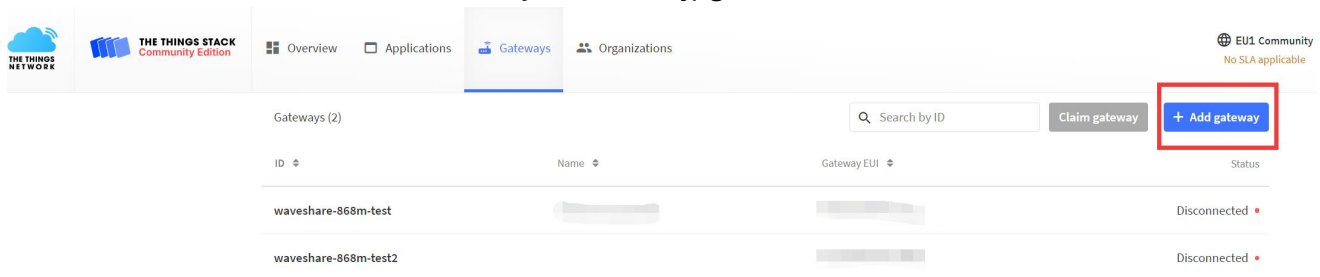
(/wiki/File:SX1302_LoRaWAN_Gateway_HAT_002.jpg)

```
pi@spi4b:~/Documents/sx1302_hal/util_chip_id $ ./chip_id
CoreCell reset through GPIO23...
SX1261 reset through GPIO23...
CoreCell power enable through GPIO22...
Opening SPI communication interface
Note: chip version is 0x10 (v1.0)
INFO: using legacy timestamp
ARB: dual demodulation disabled for all SF
INFO: found temperature sensor on port 0x39

INFO: concentrator EUI: 0x0016c[REDACTED]

Closing SPI communication interface
CoreCell reset through GPIO23...
SX1261 reset through GPIO23...
CoreCell power enable through GPIO22...
pi@spi4b:~/Documents/sx1302_hal/util_chip_id $
```

(/wiki/File: SX1302_LoRaWAN_Gateway_HAT_003.jpg)



The screenshot shows the 'Gateways' section of the The Things Stack Community Edition interface. The top navigation bar includes 'Overview', 'Applications', 'Gateways' (selected), and 'Organizations'. On the right, there is a 'EU1 Community' badge and a 'No SLA applicable' note. Below the navigation bar, there is a search bar labeled 'Search by ID' and a 'Claim gateway' button. A red box highlights the '+ Add gateway' button. Below this, a table lists two gateways:

ID	Name	Gateway EUI	Status
waveshare-868m-test	[REDACTED]	[REDACTED]	Disconnected
waveshare-868m-test2	[REDACTED]	[REDACTED]	Disconnected

(/wiki/File: SX1302_LoRaWAN_Gateway_HAT_004.jpg)

Owner ***Gateway ID** ? ***Gateway EUI** ?**Gateway name** ?**Gateway description** ?

Optional gateway description; can also be used to save notes about the gateway

Gateway Server address

The address of the Gateway Server to connect to

Require authenticated connection ?☐ Enabled

Controls whether this gateway may only connect if it uses an authenticated Basic Station or MQTT connection

Gateway status ?☒ Public

The status of this gateway may be visible to other users

Gateway location ?☒ Public

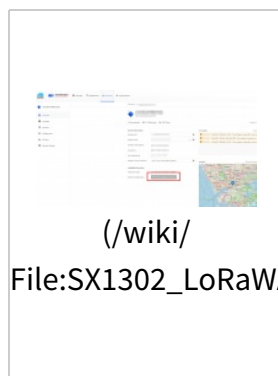
The location of this gateway may be visible to other users and on public gateway maps

Attributes ?

(/wiki/File: SX1302_LoRaWAN_Gateway_HAT_005.jpg)

- Go to the device Overview page and download the configuration file, then modify the test_conf file of sx1302_hal libraries according to the configuration file as figures.

```
cd ~/Documents/sx1302_hal/packet_forwarder/  
cp global_conf.json.sx1250.EU868 test_conf
```



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File:SX1302_LoRaWAN_Gateway_HAT_006.jpg)



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File:SX1302_LoRaWAN_Gateway_HAT_007.jpg)

Figure 1

Figure 2

- Run the packet_forwarder.

After running the codes, the gateway will keep connecting to the TTS.

```
cd ~/Documents/sx1302_hal/packet_forwarder/  
./lora_pkt_fwd -c test_conf
```

Node Access

- Please refer to Pico-LoRa-SX1262-868M (<https://www.waveshare.com/wiki/Pico-LoRa-SX1262-868M>).

Resource

Documents

- Schematic (https://files.waveshare.com/upload/e/eb/SX1302_LoRaWAN_Gateway_HAT.pdf)
- SX1302 Datasheet (https://files.waveshare.com/upload/d/df/DS_SX1302_V1.0.pdf)

Demo codes

- Semtech libraries (https://github.com/Lora-net/sx1302_hal)

Support

Technical Support

If you need technical support or have any feedback/review, please click the **Submit Now** button to submit a ticket, Our support team will check and reply to you within 1 to 2 working days. Please be patient as we

Submit Now (<https://service.waveshare.com/>)

make every effort to help you to resolve the issue.

Working Time: 9 AM - 6 PM GMT+8 (Monday to Friday)

*Retrieved from "https://www.waveshare.com/w/index.php?title=SX1302_868M_LoRaWAN_Gateway&oldid=69334
(https://www.waveshare.com/w/index.php?title=SX1302_868M_LoRaWAN_Gateway&oldid=69334)"*
